

Technical Note

Configuration Parameters for PlatoSim

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1 Abstract

This document describes the bookkeeping of the configuration parameters that are being used for PlatoSim.

2 Documents and Acronyms

2.1 Applicable Documents

AD-1: *PLATO Mission Requirements Document*, D. Agnolon

AD-2: *PLATO_01_R_BOL_i1.7*, data package v1.7 (normal camera, beginning of life, required), Anko Börner

AD-3: *PLATO_02_R_EOL_1.7*, data package v1.7 (normal camera, end of life, required), Anko Börner

AD-5: *PLATO_80_FCamB_R_EOL*, data package v1.7 (fast camera, blue, end of life, required), Anko Börner

AD-7: *PLATO_81_FCamR_R_EOL*, data package v1.7 (fast camera, red, end of life, required), Anko Börner

AD-8: PLATO-DLR-PL-TN-003, *Reference Photon Flux*

AD-9: PLATO-DLR-PL-RP-001, *Instrument Noise Budget*

AD-10: PLATO-INAF-PL-RP-001, *Telescope Optical Units Performance Analysis report*, v2

AD-11: PLATO-DLR-PL-RP-004, *Simulations and Performance Estimations*, v1

AD-12: PLATO-ESTEC-PL-RS-003, *PLATO CCD requirements*

AD-13: *Modelling CTI in PLATO-like images*, T. Prod'homme, T. Oosterbroek

AD-14: PLATO-OHV-PL-LI-0009, *PLATO Instrument Coordinate Systems*

AD-15: PLATO-KUL-PL-TN-0001, *PlatoSim Reference Frames*

AD-16: PLATO-OHB-PL-RS-005, *N_FEE User Requirements Document*, v2

AD-17: *CTI measurements, modelling, requirements*, Anko Börner

AD-18: PLATO-DLR-PL-TN-016, *Geometric Camera Model for PLATO TOU*

AD-19: *Photometry impacts of Brighter-Fatter Effect (BFE) and CCD diffusion*, Réza Samadi

AD-20: PLATO-KUL-PL-TN-003, *PLATO FEE readout modes for AIV*

2.2 Reference Documents

RD-1: *An analytical model of radiation-induced Charge Transfer Inefficiency for CCD detectors*, Short, A. et al., MNRAS 420, 3078-3085, 2013

RD-2: *Evidence for self-interaction of charge distribution in charge-coupled devices*, Guyonnet, A. et al., A & A, 575, A41, 2015

2.3 Acronyms

AR	Anti-Reflective
BFE	Brighter-Fatter Effect
BOL	Beginning Of Life
CCD	Charge-Couple Device
CVF	Charge Voltage (conversion) Factor
CTE	Charge Transfer Efficiency
CTI	Charge Transfer Inefficiency
EOL	End Of Life
FEE	Front-End Electronics
FOV	Field-Of-View
FP	Focal Plane
PIS	PLATO Image Simulator
PSF	Point Spread Function
RMS	Root Mean Square
WG	Working Group

Table 1: Configuration parameters for which input from other parties is needed.

Parameter (Sect.)	Missing information	Input required from
Sky background (5.1)	Required value in photons / pixel / s	Anko Börner (DLR)
FEE readout noise (10.2)	BOL value for fast camera	FEE WG
FEE gain (10.3)	allowed difference for detector halves	FEE WG
electronic offset (10.4)	required value at nominal CCD operating temperature	FEE WG
CCD gain (11.4)	allowed difference for detector halves	CCD WG
QE (11.5)	angle dependency	CCD WG

3 Introduction

This document was written to trace back the origin of the values for the configuration parameters in PlatoSim (developed at KU Leuven) and, by extension, other PLATO simulators, such as PIS (developed at LESIA). This allows for easy bookkeeping of the parameter values and easy comparison between the different simulators.

It is mainly based on the released system parameter tables that can be found on DLR Sharepoint under > PPT > **Simulation parameter**. Note that in v1.7 of this data package ([AD-2] and [AD-7]) information for the fast cameras is only available at EOL. If the values of the normal cameras are the same at BOL ([AD-2]) and EOL ([AD-3]), we use the EOL values ([AD-7]) for the fast cameras also for BOL.

For some of the parameters, further input is needed from other parties. This is summarised in Table 1.

For a detailed description of the individual configuration parameters, and the structure and layout of the PlatoSim configuration file, you are referred to the PlatoSim GitHub documentation pages. In the PlatoSim installation directory under /docs/tutorials, in the tutorial **ParameterReview**, you can find underlying calculations.

4 Observing Parameters

4.1 Mission Duration

The duration of the Plato mission will be **6.5 years**, as stated in requirement MRD-SC-009 in [AD-1]. Beginning-of-life (BOL) is at launch and end-of-life (EOL) is at the official end of the mission (i.e. until right after decommissioning).

The duration of the mission is used for the (linear) degradation of some effects (e.g. transmission efficiency).

4.2 Cycle Time

The cycle time is different for normal and fast cameras:

- **25 s** for the normal cameras, as stated in [AD-2] and [AD-3] (req. 300),
- and **2.5 s** for the fast cameras, as stated in [AD-5] - [AD-7] (req. 300),

This cycle time is defined by a fixed hardware sync, and is the sum of the exposure time and the duration of the readout before the next exposure starts. For the normal cameras, the latter is the total readout time; for the fast cameras, this is the duration of the frame transfer. More on the readout times can be found in Sect. 11.9.

4.3 Reference Photon Flux

For exposure time t_{exp} (expressed in seconds), the flux F_{phot} of each star is computed from its magnitude m_{λ} , the effective light-collecting area A (in cm^2), the transmission efficiency T of the optical system, the quantum efficiency of the detector Q , and the flux per second F_0 of a star with $m_{\lambda} = 0$ from

$$F_{\text{phot}} = t_{\text{exp}} \cdot F_0 \cdot T_{\lambda} \cdot Q_{\lambda} \cdot A \cdot 10^{-0.4 \cdot m_{\lambda}}, \quad (1)$$

where the λ sub-script refers to the wavelength range in which the simulation is performed. Photon flux F_0 can be computed by dividing the reference photometric flux F_{ref} by the photon energy E_p .

The value of the reference photometric flux F_{ref} , $3.6182 \cdot 10^{-12} \text{ W} \cdot \text{cm}^{-2} \cdot \mu\text{m}^{-1}$, is taken from [AD-2] - [AD-7] (req. 060). This is derived from the flux calibration, as stated in [AD-8].

Hence,

$$F_{\text{ref}} = 3.6182 \cdot 10^{-12} \frac{\text{W}}{\text{cm}^2 \cdot \mu\text{m}} \cdot \left(10^2 \frac{\text{cm}}{\text{m}}\right)^2 \cdot \left(10^{-3} \frac{\mu\text{m}}{\text{nm}}\right) \quad (2)$$

$$= 3.6182 \cdot 10^{-11} \frac{\text{W}}{\text{m}^2 \cdot \text{nm}} \quad (3)$$

$$= 3.6182 \cdot 10^{-11} \frac{\text{J}}{\text{s} \cdot \text{m}^2 \cdot \text{nm}}, \quad (4)$$

where we use the fact that $1 \text{ W} = 1 \text{ J/s}$.

At the reference wavelength λ_{ref} of 550 nm (see Sect. 8.3), the photon energy is

$$E_{\text{photon}} = \frac{h \cdot c}{\lambda_{\text{ref}}} \quad (5)$$

$$= 3.61 \cdot 10^{-19} \text{ J}, \quad (6)$$

where h and c are Planck's constant and the speed of light, respectively.

Combining Eqs. (4) and (6) gives the reference photon flux:

$$F_0 = \frac{F_{\text{ref}}}{E_{\text{photon}}} \quad (7)$$

$$= 1.00179 \cdot 10^8 \frac{\text{photons}}{\text{s} \cdot \text{m}^2 \cdot \text{nm}}, \quad (8)$$

which must be used in PlatoSim.

The derived irradiance over the Plato wavelength range can be found in Table 2-1 in [AD-8] and is shown in Fig. 1. Integrated for Plato wavelength range (450 - 1050 nm, according to reqs. 240 and 250 in [AD-2] - [AD-7]) at magnitude 0, the irradiance is $5.327 \cdot 10^6 \text{ photons} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}$. This is the value that must be used in PIS and can also be obtained by multiplying the photon flux with the throughput bandwidth (see Sect. 8.4).

5 Sky Parameters

5.1 Sky Background

At this moment, the required background signal is expressed in $\text{e}^-/\text{pixel}/\text{s}$, which would result in different background levels in $\text{photons}/\text{pixel}/\text{s}$ for BOL and EOL, which is counterintuitive. So rather than using the required value of $100 \text{ e}^-/\text{pixel}/\text{s}$ (req. 160 in [AD-2] - [AD-7]), we decided to use the required value from an older version of these files, namely $342 \text{ photons}/\text{pixel}/\text{s}$.

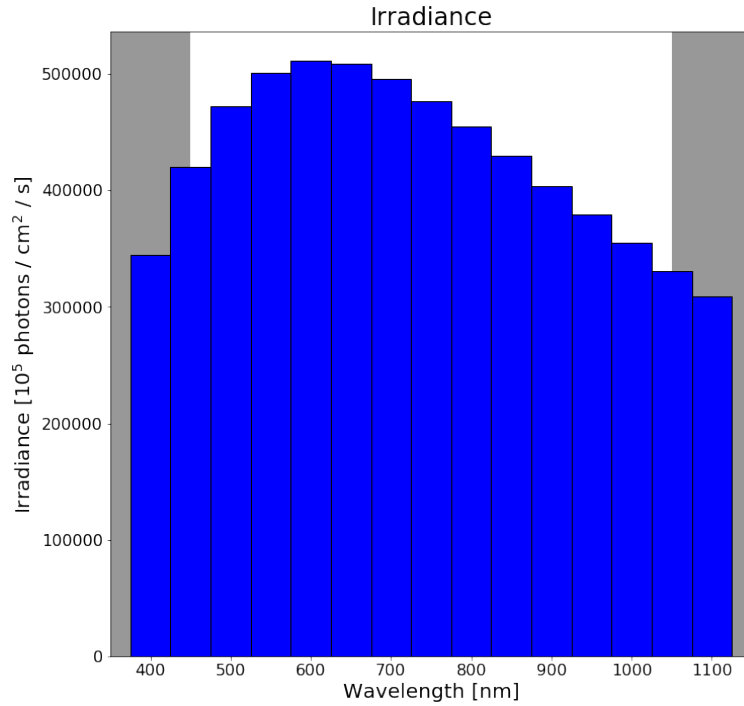


Figure 1: Irradiances at magnitude 0 for a G0V star with 6000 K, as listed in Table 2-1 in [AD-8]. These are the photon densities from a target stars, assuming 15 bins of 50 nm wide, between 400 and 1100 nm. The regions marked in gray indicate what falls outside the PLATO wavelength range.

5.2 Cosmics

According to req. 790 in [AD-2] - [AD-7], the cosmic hit rate amounts to **10 events/cm²/s**.

Note that the image area is susceptible to cosmic hits during exposure and readout, the smearing map (i.e. parallel over-scan) during readout, and the bias register map (i.e. parallel pre-scan) only during the time required to read out the register.

For fast cameras, the shield protecting the lower half of the CCD from incoming radiation, does not stop cosmics from hitting the detector.

6 Platform Parameters

6.1 Orientation of the Solar Panels

The orientation angle of the solar panels is the roll angle of the platform, which enables orienting the solar panels away from the Sun each quarter. By convention, the roll angle must be set to **0°** at the start of the first quarter and increased by **90°** at the beginning of each quarter.

6.2 Jitter File

At this stage, the following jitter files, delivered by industry, are available:

- AOCs_NM_pointing_7h.txt,
- PlatoJitter_Airbus.txt,

- PlatoJitter_OHB.txt,
- PlatoJitter_OHB_june2018.txt,
- PlatoJitter_Thales.txt,
- and PlatoJitter_Thales_detrended.txt.

6.3 Jitter Parameters

The values of the parameters to generate the spacecraft jitter ourselves, as a random walk (Brownian motion),

- $\text{RMS}(\text{yaw}) = \text{RMS}(\text{pitch}) = \mathbf{0.04 \text{ arcsec}}$,
- $\text{RMS}(\text{roll}) = \mathbf{0.08 \text{ arcsec}}$,
- and a timescale of **3600 s**

are pure test values and are not based on requirements.

7 Telescope Parameters

7.1 Light Collecting Area

The value for the light collecting area of the telescope, **113.1 cm^2** , is taken from [AD-2] - [AD-7] (req. 410), where it is referred to as the effective aperture of the telescope. This corresponds to an effective pupil size diameter of 12 cm (req. 400 in [AD-2] - [AD-7]).

7.2 Transmission Efficiency

The transmissivity of the optics is described by the files `optics_transmission_1.2pc_bol_cutoff_19-03-31.txt` and `optics_transmission_1.2pc_eol_cutoff_19-03-31.txt` (see req. 480 in [AD-2] - [AD-7]). A plot is shown in Fig. 2. In these curves the anti-reflective (AR) coating of 1.2% is included.

Taking the average over the wavelength range 450 - 1050 nm, yields an average transmissivity of **0.8135** at BOL and **0.7945** at EOL.

7.3 Drift Parameters

The values of the parameters to generate the thermo-elastic drift ourselves, as a random walk (Brownian motion),

- $\text{RMS}(\text{yaw}) = \text{RMS}(\text{pitch}) = \text{RMS}(\text{roll}) = \mathbf{2 \text{ arcsec}}$
- and a timescale of **86400 s**

are pure test values and are not based on requirements.

8 Camera Parameters

8.1 Plate Scale

The value of the plate scale used in PlatoSim, **$0.8333 \text{ arcsec}/\mu\text{m}$** , is the ratio of the plate scale of the optics (15 arcsec/pixel according to req. 440 in [AD-2] - [AD-7]) and the pixel size (in square) of the CCD ($18 \mu\text{m/pixel}$ according to req. 630 of [AD-2] - [AD-7]).

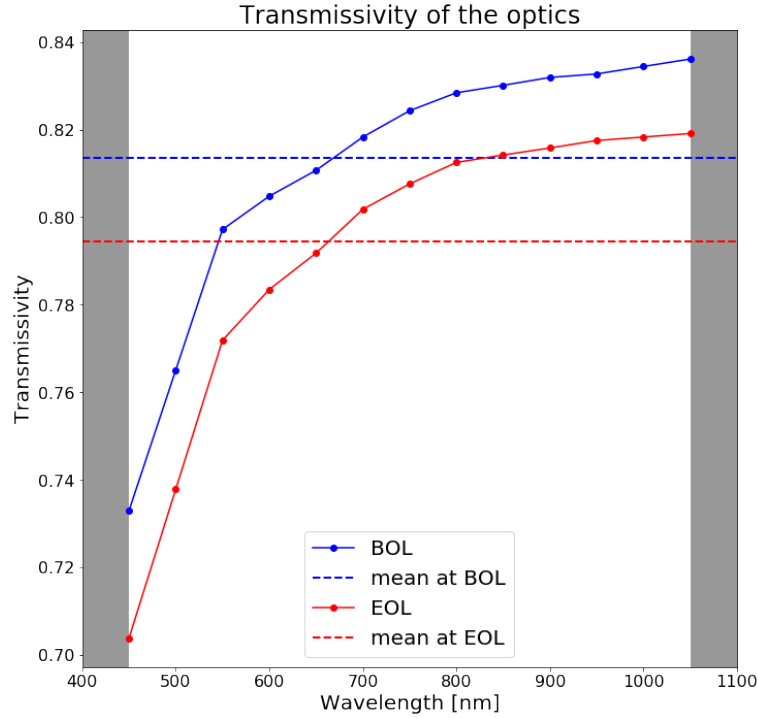


Figure 2: Transmissivity of the optics (incl. AR coatings), as specified by req. 480 in [AD-2] - [AD-7]. The regions marked in gray indicate what falls outside the PLATO wavelength range.

8.2 Focal Length

The value of the focal length, **0.24752 m**, is taken from [AD-2] - [AD-7] (req. 420). This value is obtained as

$$\text{focal length} = \frac{\text{pixel size}}{\tan\left(\frac{\text{plate scale}}{2}\right)}. \quad (9)$$

and slightly differs from the value from the Zemax model (0.24712595 m).

Expressed in pixels, the focal length can be calculated as

$$\text{focal length} = \frac{1}{\tan(\text{plate scale})}, \quad (10)$$

which yields a value of **13751 pixels**.

8.3 Reference Wavelength

The reference photometric wavelength or throughput wavelength λ_{ref} is **550 nm**, as described in [AD-2] - [AD-7] (req. 070).

8.4 Throughput Bandwidth

The value of the throughput bandwidth $\Delta\lambda_{\text{ref}}$ is obtained from the reference photon flux and the irradiance integrated for the Plato wavelengths at magnitude 0 (see Sect. 4.3):

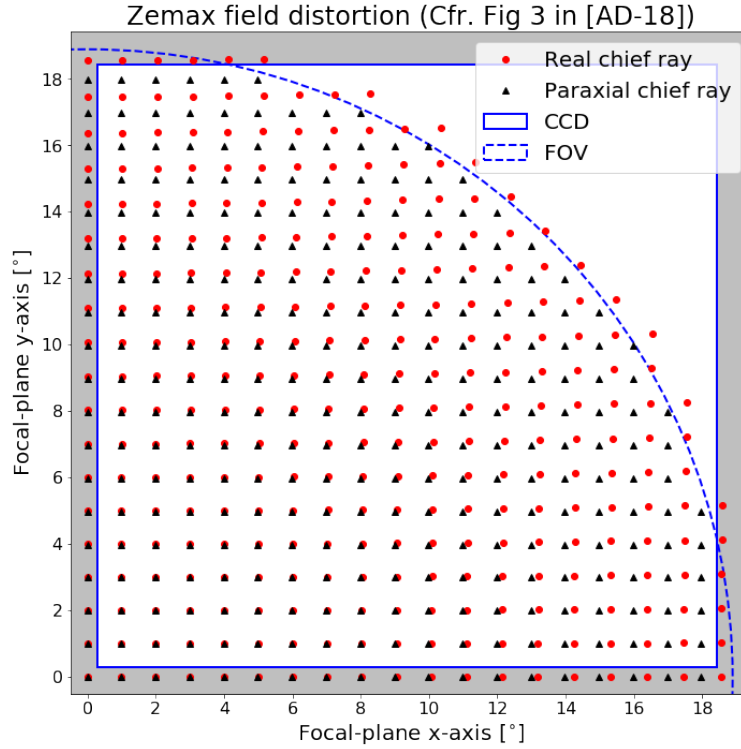


Figure 3: Zemax field distortion (cfr. Fig. 3 in AD-18) over one quadrant of the FPA, in steps of 1° . The paraxial chief ray coordinates are represented by black triangles, the real chief ray coordinates by red dots. The gray region indicates what falls off the CCD, the blue dashed line indicates the size of the FOV.

$$\Delta\lambda_{\text{ref}} = \frac{\text{irradiance}}{\text{reference photon flux}} \quad (11)$$

$$= 532 \text{ nm} \quad (12)$$

8.5 Aberration Correction

The calculation of the aberration correction is based on a circular Earth orbit around the Sun and does not take the Lissajous orbit of the satellite around L2 into account. The aberration correction for the spacecraft pointing is taken into account.

8.6 Field Distortion

The increase in image magnification with the distance from the optical axis is characterised by a radial distortion model which considers barrel distortion. This model assumes the distortion to depend only on the field radius and not on the position in the FOV. The effect of field distortion is shown in Fig. 3.

The distortion, defined as the difference in radial distance to the optical axis between the distorted and undistorted coordinates (r_d and r resp.), can then be expressed in terms of the undistorted radial distance r as follows:

$$r_d - r = r \cdot (k_1 \cdot r^2 + k_2 \cdot r^4 + k_3 \cdot r^6), \quad (13)$$

Table 2: Field distortion coefficients and their inverse, as derived from a polynomial fit on the distortion data provided by Demetrio Magrin.

	k_1	k_2	k_3
Eq. (13)	0.316257210577	0.066373219688	0.372589221219
Eq. (14)	-0.317143032936	0.242638513347	-0.459260203502

where the higher-order terms are neglected. Here r and r_d are expressed in normalised camera coordinates (i.e. pixel coordinates in the focal plane, divided by the focal length expressed in pixels).

Inversely, the distortion can also be expressed in terms of the distorted radial distance r_d as follows:

$$r - r_d = r_d \cdot (k_1 \cdot r_d^2 + k_2 \cdot r_d^4 + k_3 \cdot r_d^6), \quad (14)$$

but note that the coefficients (k_1, k_2, k_3) have different values in Eqs. (13) and (14). The meaning and units of r and r_d stay the same.

The paraxial and real chief ray coordinates (from which to calculate r and r_d resp.) are tabulated in a file provided by Demetrio Magrin in an email to Joris De Ridder on July 1st, 2015. The coefficients (k_1, k_2, k_3) in Eqs. (13) and (14) are derived from a model fit using the Levenberg-Marquardt algorithm on the distortion data and are tabulated in Table 2.

9 PSF Parameters

9.1 Demetrio's PSFs

The pre-computed PSFs, provided by Demetrio Magrin, are described in [AD-10]. These monochromatic PSFs are associated with the optical design from June 2015. They have a resolution of 1/128 pixels and a physical size of 8×8 pixels.

9.2 Gaussian PSF

The fraction of the energy of a circular Gaussian PSF, enclosed in a box of dimensions $2n\sigma \times 2n\sigma$ pixels is

$$\left(\operatorname{erf} \left(\frac{n}{\sqrt{2}} \right) \right)^2,$$

where erf denotes the error function and σ the width of the Gaussian (in pixels) in both directions.

According to reqs. 450 and 460 in [AD-2] - [AD-7], 32% of the energy should be ensquared in 1×1 pixel. This means that:

$$2n\sigma = 1 \text{ pixel} \quad (15)$$

and

$$\left(\operatorname{erf} \left(\frac{n}{\sqrt{2}} \right) \right)^2 = 0.32, \quad (16)$$

which can be combined to

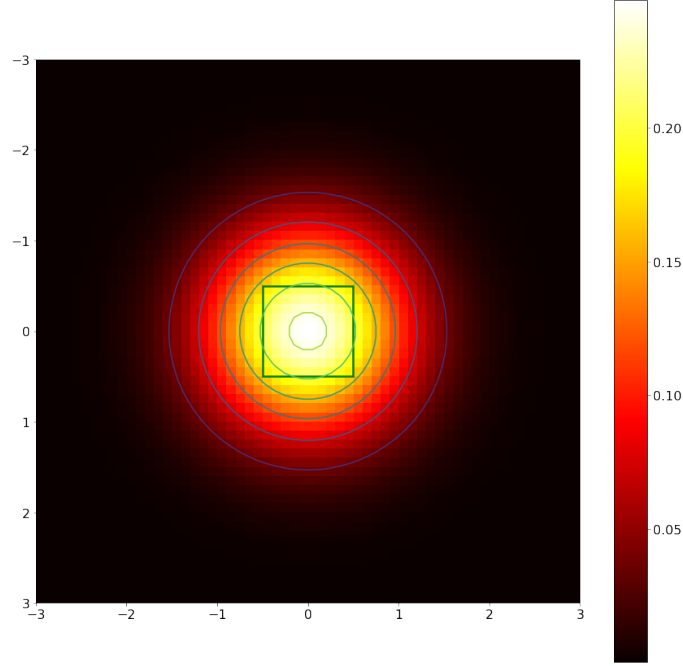


Figure 4: Gaussian PSF with a width of 0.639 pixels, with a 1×1 pixel box overplotted, ensquaring 32% of the PSF's energy.

$$n = \sqrt{2} \cdot \operatorname{erf}^{-1}(\sqrt{0.32}) = \frac{1 \text{ pixel}}{2\sigma}, \quad (17)$$

and thus

$$\sigma = \frac{1 \text{ pixel}}{2 \cdot \sqrt{2} \cdot \operatorname{erf}^{-1}(\sqrt{0.32})}. \quad (18)$$

The required width of the Gaussian PSF is therefore **0.639 pixels**.

9.3 Analytic PSF

The principle of the analytic non-Gaussian PSFs, provided by Carsten Paproth, is described in Sect. 2.5 of [AD-11]. The final set of parameters are a result of an extensive optimisation procedure (requiring a lot of manual work).

9.4 Charge Diffusion and Jitter Smoothing

Charge diffusion and jitter smoothing are both implemented as a convolution with a Gaussian diffusion kernel. For the charge diffusion we use a kernel width of **0.2 pixels** (estimate from old CCD measurement; priv. comm. with Carsten Paproth). The kernel for the jitter smoothing has a width of **0.5 sub-pixels**.

10 Front-End Electronics Parameters

10.1 Nominal Operating Temperature

The nominal operating temperature of the FEE of -63°C is taken from [AD-16].

10.2 Readout Noise

The readout noise for the FEE for the normal cameras is $32.0\text{ e}^{-}/\text{pixel}$ at BOL and $37.0\text{ e}^{-}/\text{pixel}$ at EOL, which is taken from [AD-2] and [AD-3] (req. 860).

For the fast cameras, the FEE readout noise at EOL is as much as $200\text{ e}^{-}/\text{pixel}$ according to req. 860 in [AD-5] - [AD-7]. Until further notice, we advise to use this value for BOL as well.

10.3 Gain

The total ADC gain, g_{total} , of $25\text{ e}^{-}/\text{ADU}$ for the normal cameras (req. 880 in [AD-2] and [AD-3]) and $15\text{ e}^{-}/\text{ADU}$ for the fast cameras (req. 880 in [AD-5] - [AD-7]) is split into:

- CCD gain, g_{CCD} , (expressed in $\mu\text{V}/\text{e}^{-}$) for the conversion from e^{-} to μV (see Sect. 11.4),
- and FEE gain, g_{FEE} , (expressed in $\text{ADU}/\mu\text{V}$) for the conversion from μV to e^{-} .

The total gain can then be written as

$$g_{\text{total}}[\text{e}^{-}/\text{ADU}] = \left(\frac{1}{g_{\text{CCD}}[\mu\text{V}/\text{e}^{-}]} \right) \cdot \left(\frac{1}{g_{\text{FEE}}[\text{ADU}/\mu\text{V}]} \right) \quad (19)$$

$$= \left(\frac{1}{1.8\mu\text{V}/\text{e}^{-}} \right) \cdot \left(\frac{1}{g_{\text{FEE}}[\text{ADU}/\mu\text{V}]} \right) \quad (20)$$

Hence can the FEE gain be computed as:

$$g_{\text{FEE}}[\text{ADU}/\mu\text{V}] = \frac{1}{g_{\text{CCD}}[\mu\text{V}/\text{e}^{-}] \cdot g_{\text{total}}[\text{e}^{-}/\text{ADU}]} \quad (21)$$

For the normal cameras this results in an FEE gain of $0.0222\text{ ADU}/\mu\text{V}$, for the fast cameras in a gain of $0.0370\text{ ADU}/\mu\text{V}$.

PlatoSim allows for a difference in FEE gain for both detector halves. The difference between the two should not exceed a configurable threshold, but this has not been characterised yet.

The temperature stability (req. 840 in [AD-2] - [AD-7]) is $300\text{ ppm}/\text{K}$, or $300 \cdot 10^{-6}\text{ ADU}/\mu\text{V}/\text{K}$.

10.4 Electronic Offset

Currently there is no requirement for the value of the electronic offset. We've adopted a value of 1000 ADU .

According to req. 850 in [AD-2] - [AD-7] the temperature stability of the electronic offset is $1\text{ ADU}/\text{K}$ for all cameras.

11 CCD Parameters

11.1 Nominal Operating Temperature

The nominal operating temperature of the CCD, -70°C , is taken from [AD-12] (req. CCD-GEN-008).

11.2 Pixel Size

The pixel size is **18 μm** (in square), according to [AD-2] - [AD-7] (req. 630).

11.3 Brighter-Fatter Effect

The implementation of the brighter-fatter effect is based on [RD-2]. The values for all characterising parameters are taken from [AD-19].

Each pixel is affected by all pixels that are not further away than **8 pixels** in the row and column direction. Parameters p_0 and p_1 in Eq. (18) in [RD-2] have values **0.05** and **0.15** at a reference flux of **10^6 e^-** .

11.4 Gain

The CCD gain (at the nominal operating temperature of the CCD) is listed in [AD-2] - [AD-7] as the charge-voltage conversion factor (CVF) (req. 680), for which the value is **$1.8 \mu\text{V}/\text{e}^-$** .

PlatoSim allows for a difference in CCD gain for both detector halves. The difference between the two should not exceed a configurable threshold, but this has not been characterised yet.

The temperature dependency of the CVF is discussed in Sect. 5.2.2.5 of [AD-9]. The temperature stability of the CCD gain for temperature changes w.r.t. the nominal operating temperature (i.e. the temperature dependent CVF error ΔCVF) is **$-0.001 \mu\text{V}/\text{e}^-/\text{K}$** .

11.5 Quantum Efficiency

The quantum efficiency Q is described by a function

$$Q = Q_0 \cdot f(\lambda) \cdot g(\theta), \quad (22)$$

where λ denotes the wavelength dependency and θ the angle dependency.

In the file `ccd_qe_min_bol_19-03-31.txt`, mentioned in req. 650 in [AD-2] - [AD-7], the values for $Q_0 \cdot f(\lambda)$ are tabulated for wavelengths between 400 and 1000 nm. These values are plotted in Fig. 5. The mean quantum efficiency, E_{QE} , obtained by taking the average of $Q_0 \cdot f(\lambda)$ over the wavelength range between 450 and 1000 nm, is **0.6218**.

The mean efficiency caused by the angle dependency of the quantum efficiency, $E_{\text{ang.}}$, is **1.01** according to req. 660 in [AD-2] - [AD-7]. No additional information regarding variations over the FOV are currently available. The mean efficiency caused by the angle dependency will be applied on the entire FOV, until further notice.

11.6 Full-Well Saturation (Blooming)

The value for the full-well saturation limit is **$900000 \text{ e}^-/\text{pixel}$** , as stated in req. 640 of [AD-2] - [AD-7].

11.7 Digital Saturation

According to req. 870 of [AD-2] and [AD-7], the ADC quantisation is **16 bits** for the all cameras. This corresponds to a digital saturation limit of **$65535 \text{ ADU}/\text{pixel}$** .

11.8 Readout Noise

The CCD readout noise is mentioned in req. 780 in [AD-2] - [AD-7]. It is **38.7 e^-** at BOL and increases to **44.3 e^-** at EOL.

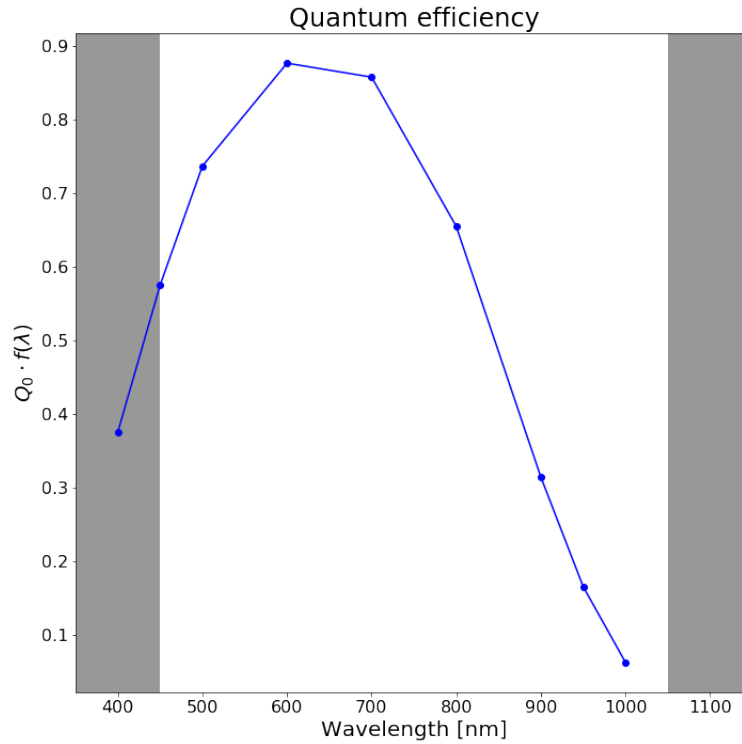


Figure 5: Terms $Q_0 \cdot f(\lambda)$ from Eq. (22), as tabulated in the file `ccd_qe_min_bol.txt`, mentioned in req. 650 in [AD-2] - [AD-7]. The regions marked in gray indicate what falls outside the PLATO wavelength range.

11.9 Readout Time

A distinction is made between the time required to transfer charges one row down (towards the readout register) in case the readout register is read out, or not. The former is **110 μ s**, the latter **90 μ s**, as taken from [AD-20]. The difference between the two is due to two delay parameters recommended by Teledyne e2v for clock settling. Settling times are to ensure one clock has reached its low level before another clock starts to rise. It ensures good charge transfer, as without it you could cause charge to be lost. This is particularly the case when dealing with the (parallel) transfer into the readout register, as the rise and fall times of the image clocks are a lot slower than those of the register clocks.

The time needed to transfer the content of the readout register by one pixel (towards the output node) is **340 ns**, which is also taken from [AD-20]. Note that the left and right detector half are read out simultaneously.

11.10 Readout Mode

Both the normal and the fast camera can be operated in **nominal-readout** mode (full frame for the normal camera, frame transfer for the fast camera) or in **partial-readout** mode. In nominal-readout mode, all rows in the image area are read out from the readout register. In partial-readout mode, only a limited number of rows in the image area are read out from the readout register. The other rows in the image area will be dumped.

11.11 Flatfield Noise RMS

The flatfield noise RMS is listed in [AD-2] - [AD-7] (req. 730) as the local PRNU and amounts to **0.010**.

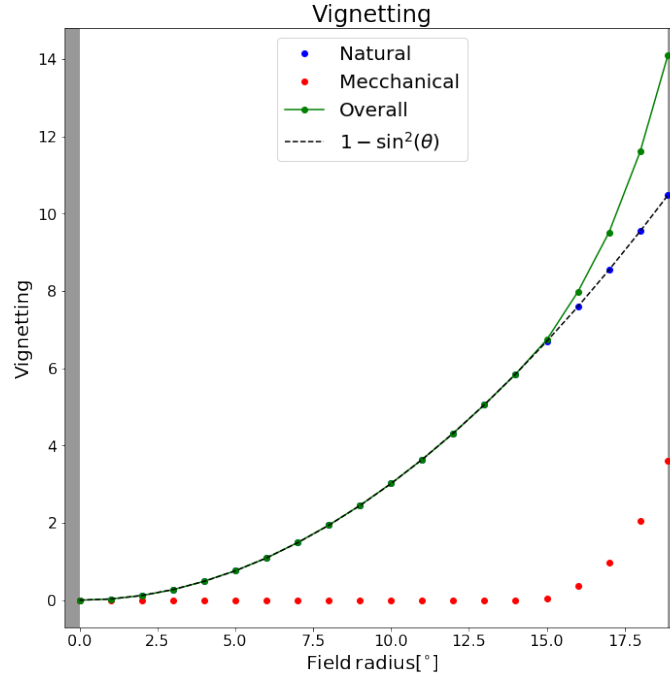


Figure 6: Natural, mechanical, and overall vignetting as function of the field radial coordinate (see [AD-10]). The regions marked in gray indicate what falls outside the FOV.

11.12 Vignetting

Vignetting, i.e. the brightness attenuation towards the edges of the FOV, depends on the incident angle or line of sight of a pixel. The expected efficiency due to (natural and mechanical) vignetting (see Fig. 6), i.e. the mean over all pixels of one camera, is **0.904** at BOL and **0.902** at EOL according to req. 520 in [AD-2] - [AD-7], which corresponds to a loss of **18%** at the edge of the FOV (req. 515).

The size of the FOV can be derived from req. 432 in [AD-2] - [AD-7], stating that the FOV should measure 1111 degrees². This corresponds to the surface of the spherical cap indicated in gray on Fig. 7. From the formula for the surface area of a spherical cap

$$A = 2\pi R^2(1 - \cos \theta) \quad (23)$$

$$= 1111 \text{ degrees}^2 \quad (24)$$

in a sphere with radius $R = \frac{360^\circ}{2\pi}$, we can extract the size of the FOV:

$$\theta = \arccos\left(\frac{-A}{2\pi R^2} + 1\right) \quad (25)$$

$$= \arccos\left(\frac{-1111 \text{ degrees}^2}{2\pi \left(\frac{360^\circ}{2\pi}\right)^2} + 1\right), \quad (26)$$

yielding a FOV of **18.8908°**.

11.12.1 Natural Vignetting

The efficiency caused by natural vignetting of the optics for pixel (*row, column*) can be approximated by

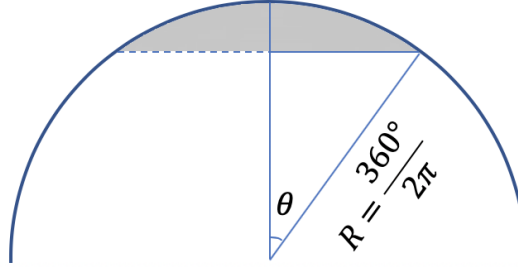


Figure 7: Sphere representing the sky, which is 360° all around. The radius of this sphere is hence $R = \frac{360^\circ}{2\pi}$. The surface of the FOV corresponds to the spherical cap marked in gray. The size of the FOV is indicated by the angle θ .

$$E_{\text{vig.}}(\text{row}, \text{column}) = \cos^2(\theta(\text{row}, \text{column})), \quad (27)$$

where $\theta(\text{row}, \text{column})$ is the incident angle or line of sight for that pixel.

11.12.2 Mechanical Vignetting

Mechanical vignetting has been implemented as if all incoming radiation (apart from the cosmic hits) is shielded off beyond the FOV.

11.13 Polarisation

Optical elements lead to partial polarisation of the light of target star, which means there are preferred directions for light propagation. Though this effect depends on the incident angle or line of sight of a pixel, no information is available about this angle dependency. Therefore we will apply the expected efficiency of **0.989** (req. 530 of [AD-2] - [AD-7]) of the entire FOV, until further notice.

11.14 Particulate Contamination

The efficiency due to particulate contamination (i.e. the unintended presence of particles on (mostly optical) surfaces) is **0.972**, as listed in req. 500 of [AD-2] - [AD-7] .

11.15 Molecular Contamination

The efficiency due to molecular contamination (mostly caused by outgassing of materials in the first phase of the mission) is **0.9573**, as listed in req. 510 of [AD-2] - [AD-7] .

11.16 Dark Signal

Dark signal is the relatively small electric current that is generated in the CCD when no outside radiation is entering the device, arising from thermal energy within the silicon lattice of the CCD.

According to reqs. 700 and 710 of [AD-2] - [AD-7], the dark current increases from **$1.2 \text{ e}^-/\text{s/pixel}$** at BOL to **$4.5 \text{ e}^-/\text{s/pixel}$** at EOL, with a temperature dependency of **$5 \text{ e}^-/\text{s/pixel/K}$** .

The dark-signal non-uniformity (DSNU), a systematic (fixed-pattern) deviation of a pixels's dark current w.r.t its nominal value, is **15 %**, as reported in Sect. 5.1.1.3 of [AD-9].

Table 3: CTI values for parallel and serial transfers, and the expected efficiency caused by CTI for the different cameras at BOL.

Camera	CTI_{parallel}	CTI_{serial}	E_{CTI}
normal	$5.42 \cdot 10^{-6}$	$5.42 \cdot 10^{-7}$	0.987
fast (blue)	$8.13 \cdot 10^{-6}$	$8.13 \cdot 10^{-7}$	0.981
fast (red)	$8.43 \cdot 10^{-6}$	$8.43 \cdot 10^{-7}$	0.981

Note that these values only apply to operational and not to ambient conditions. At room temperature we advise to use the equations from Sect. 2 in [AD-20] to calculate the dark current:

$$\text{dark signal} = A \cdot T^{3/2} \cdot \exp\left(\frac{-E_g(T)}{2kT}\right), \quad (28)$$

with (room) temperature T expressed in K, k the Boltzmann constant expressed in eV/K, and

$$A = 3.816788 \cdot 10^{10}, \quad (29)$$

$$E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta}, \quad (30)$$

$$E_g(0) = 1.166, \quad (31)$$

$$\alpha = 4.74 \cdot 10^{-4}, \quad (32)$$

$$\beta = 636. \quad (33)$$

Since the implemented temperature dependency does not apply to this temperature regime, we advise to set the temperature stability to zero in that case.

11.17 Charge Transfer Inefficiency (CTI)

11.17.1 Simple

When moving charges down towards and into the readout register (parallel transfer) and within the register (serial transfer), some charge stays behind and is released at a later point in time. This loss in efficiency is characterised by the CTI, which is different for parallel and serial transfers (i.e. we distinguish between CTI_{parallel} and CTI_{serial}).

Since the left- and right-hand side of the CCDs are read out separately (with their own readout register and electronics), the efficiency caused by CTI for pixel ($row, column$) due to CTI is:

$$E_{\text{CTI}}(row, column) = (1 - CTI_{\text{parallel}})^{\text{row}} \cdot (1 - CTI_{\text{serial}})^{\max\{column, 4510 - column + 1\}}. \quad (34)$$

Req. 762 in [AD-2] - [AD-7] states that:

$$CTI_{\text{serial}} = 0.1 \cdot CTI_{\text{parallel}}. \quad (35)$$

Tables 3 and 4 list values for CTI_{parallel} and CTI_{serial} , together with the expected efficiency caused by CTI. Since this simple approach to CTI has been abandoned for a more sophisticated model (see [AD-17]; CTI depending on radiation level and signal intensity), mimicking the results of Short et al. (see Sect. 11.17.2), these parameters no longer occur in [AD-2] - [AD-7]. We therefore took these values from an older version of these files (reqs. 750, 760, and 770).

Table 4: CTI values for parallel and serial transfers, and the expected efficiency caused by CTI for the different cameras at EOL.

Camera	CTI_{parallel}	CTI_{serial}	E_{CTI}
normal	$4.97 \cdot 10^{-5}$	$4.97 \cdot 10^{-6}$	0.891
fast (blue)	$7.48 \cdot 10^{-5}$	$7.48 \cdot 10^{-6}$	0.842
fast (red)	$7.70 \cdot 10^{-5}$	$7.70 \cdot 10^{-6}$	0.838

Table 5: Characteristics of the four trap species used for Plato, according to [AD-13].

Trap species	Release time constant (at 203 K)	Trap density [traps/pixel]	Capture cross-section [m^{-2}]
fast	$2.37 \cdot 10^{-4}$	9.80	$2.46 \cdot 10^{-20}$
intermediate	$2.43 \cdot 10^{-2}$	3.31	$1.74 \cdot 10^{-22}$
intermediate	$2.03 \cdot 10^{-3}$	1.56	$7.05 \cdot 10^{-23}$
slow	$1.40 \cdot 10^{-1}$	13.24	$2.45 \cdot 10^{-23}$

11.17.2 Short et al. 2013

The parameters for the implementation of the Short et al. 2013 algorithm [RD-1] for CTI are discussed in [AD-13]. We use **4 trap species**, the characteristics of which are listed in Table 5. These values are valid for a temperature of **203 K**. The optimal value for the β parameter is **0.37** also according to [AD-13].

12 Camera Groups

The normal cameras are divided into four groups, plus the two fast cameras, all of which have their own tilt and azimuth angle, as specified in Table 7-1 in [AD-14]. The tilt angle is also listed in req. 280 in [AD-2] - [AD-7].

13 CCD Positions

Each camera comprises four CCDs as depicted in Fig. 8. The characterising parameters of the CCD positions are summarised in Table 7, which corresponds to Table 1 in [AD-15].

The number of rows and columns, N_{rows} and N_{columns} are specified in reqs. 610 and 620 in [AD-2] - [AD-7]. The normal cameras use full-frame CCDs, whereas the fast cameras use frame transfer CCD for which only the upper half is being illuminated during exposure (req. 600), explaining the values for r_{exp} .

The values for Δx_{CCD} come from the butting distance being 2.6 mm, according to req. 285 in [AD-2] - [AD-7].

Table 6: Tilt and azimuth angles for the four telescope/camera groups, plus the two fast cameras, according to Table 7-1 in [AD-14].

Group	Tilt angle [$^{\circ}$]	Azimuth angle [$^{\circ}$]
1	9.2	45.0
2	9.2	135.0
3	9.2	-135.0
4	9.2	-45.0
Fast	0.0	0.0

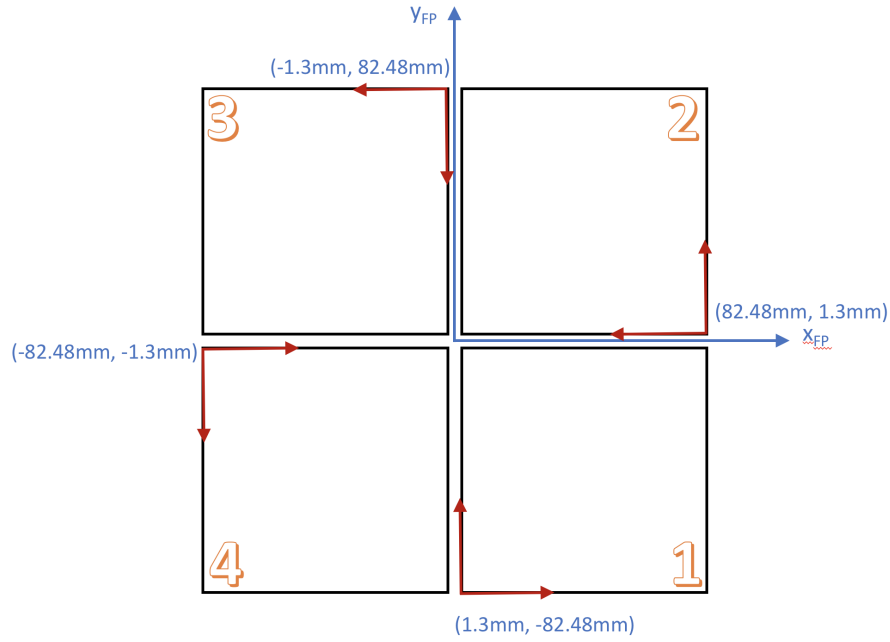


Figure 8: Schematic overview of the four CCDs of a telescope. The blue coordinate system (x_{FP}, y_{FP}) is the focal-plane reference frame, with origin at the optical axis. The red coordinate systems are the CCD reference frames of the four CCDs. The coordinates in blue are the focal-plane coordinates of the origin of the CCD reference frames. The butting distance of 2.6 mm is the distance between the CCDs.

Taking this butting distance, the CCD dimensions, and the pixel size (see Sect. 11.2) into account, the values for Δy_{CCD} can be obtained.

Table 7: Characteristics for the CCD position of the four CCDs of a normal camera (1, 2, 3, and 4) and of a fast camera (1(F), 2(F), 3(F), and 4(F)), as adopted from Table 1 in [AD-15]. The first column lists the CCD codes: 1, 2, 3, and 4 for a normal camera, 1(F), 2(F), 3(F), and (F) for a fast camera. N_{rows} and $N_{columns}$ are the number of rows and columns of the CCD, r_{exp} is the first row that is exposed; for the fast camera only rows 2255 until 4510 are exposed. Δx_{CCD} and Δy_{CCD} are the coordinates of the FP origin in the different CCD reference frames; γ_{CCD} is the CCD orientation angle.

CCD	N_{rows}	N_{cols}	r_{exp}	Δx_{CCD} (mm)	Δy_{CCD} (mm)	γ_{CCD} (deg)
1	4510	4510	0	-1.3	82.48	0
2	4510	4510	0	-1.3	82.48	90
3	4510	4510	0	-1.3	82.48	180
4	4510	4510	0	-1.3	82.48	270
1 (F)	4510	4510	2255	-1.3	82.48	0
2 (F)	4510	4510	2255	-1.3	82.48	90
3 (F)	4510	4510	2255	-1.3	82.48	180
4 (F)	4510	4510	2255	-1.3	82.48	270